Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Correct: C

2. (2 points) An LED’s forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always 0.7 V

B. Always zero

C. Equal to any source

D. Device- and condition-dependent

Correct: D

3. (2 points) [State/National] KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Charge conservation

B. Equal current in all branches

C. Energy creation

D. Resistance addition only

Correct: A

4. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Correct: C

5. (2 points) A series resistor mainly…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Creates charge

B. Guarantees any voltage is safe

C. Reverses polarity

D. Limits current

Correct: D

6. (2 points) [State/National] Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Opposite the assumed direction

B. Impossible circuit necessarily

C. Negative charge disappears

D. No voltage exists

Correct: A

7. (2 points) At fixed R, doubling V changes power by…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. A factor of one half

B. No change

C. A factor of four

D. A factor of two

Correct: C

8. (2 points) A diode is generally…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. An ideal fixed resistor

B. Always conducting both ways

C. A mechanical switch

D. Nonlinear

Correct: D

9. (2 points) [State/National] Resistor dissipation is…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. I^2R

B. Always negative

C. Independent of current

D. Equal to charge

Correct: A

10. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Steady charge flow conserves charge while energy changes form. (4 points).

11. (2 points) A 7 V ideal battery drives a 3.5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 3 and 5 ohms. Find equivalent resistance.

answer: 8 ohms (2 points).

Numerical target 8; absolute tolerance 0.016. Accepted units: , ohms.

13. (2 points) Two identical 6 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 3 ohms (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 7 V. Find power.

answer: 14 W (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , W.

15. (2 points) [State/National] A node has incoming currents 4 A and 2 A and one outgoing current. Find the outgoing current.

answer: 6 A (2 points).

Numerical target 6; absolute tolerance 0.012. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) Describe a diagnostic use of a ground-fault circuit interrupter without claiming it replaces every protection device.

answer: It detects a current imbalance between outgoing and returning paths and interrupts the circuit; overcurrent and other hazards may require additional protection. (2 points).

19. (2 points) [State/National] A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).